# Predicting Singapore HDB Rental Prices under Temporal Distribution Shift

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***Abstract*—Predicting Singapore HDB rents requires accounting for changing market conditions as well as property and location characteristics. We analysed 150,000 labelled rental records from January 2021 to March 2025 and developed features for 50,000 future test records. Exploratory analysis motivated category normalisation, invalid-source masking and chronological validation. We compared record-handling policies, Ridge regression, CatBoost and a multilayer perceptron using two earlier validation windows for selection. The selected CatBoost model retained all training records and used 35 core, HDB and MRT predictors. On 25,984 later observations, it achieved RMSE 458.740 SGD and MAE 334.412 SGD, reducing RMSE by 21.21% relative to Ridge. Its mean underprediction was 13.551 SGD. School, mall and historical-rent additions did not meet our minimum improvement threshold. A 16-month diagnostic also favoured CatBoost, although its observations overlapped the other windows. The findings support nonlinear modelling and careful temporal evaluation, while snapshot amenities, subgroup errors and unavailable future labels limit conclusions about deployment accuracy.**

***Index Terms*—HDB rental prediction, temporal validation, feature engineering, regression, spatial auxiliary data**

## I. Introduction

### A. Motivation and Problem Context

Accommodation costs place a substantial financial burden on households in Singapore. As the course overview emphasises, renting can be a practical necessity for people who are unable or ineligible to purchase housing, including singles, young couples waiting for a new flat, and foreign workers [1]. Renters need to judge what their budget can secure and compare asking rents across properties; landlords and real estate agents need realistic pricing references. These needs motivate our study of Housing & Development Board (HDB) flats. Given a flat's characteristics and approval month, we predict its monthly rent from historical observations and assess the reliability of predictions for later periods.

This task presents three challenges. First, rental levels change over time: in the supplied training data, mean rent rises from SGD 2,121 in 2021 to SGD 3,036 in 2023, followed by much smaller changes in 2024 and early 2025. These descriptive averages also reflect the mix of properties observed, so they do not isolate a causal market effect. Nevertheless, the pattern makes continued extrapolation of an earlier growth trend questionable. Second, location and property characteristics are interdependent: town, street, flat type and floor area may encode overlapping information, while amenity features may contribute differently across models. Third, repeated feature vectors and inconsistent derived attributes require careful treatment. Without identifiers for individual flats or contracts, matching rows cannot be assumed to represent duplicate transactions.

Useful rental estimates must remain credible when the prediction period extends beyond the observed market conditions. An underestimated rent can mislead a tenant's budget, while an overestimated rent can distort property comparisons or pricing expectations. Moreover, low average error can conceal poor performance for uncommon flat types or less represented locations. We therefore consider temporal calibration and subgroup errors alongside overall accuracy. This framing treats the model as a reference based on observable attributes, rather than a complete valuation of an individual flat's condition or contractual terms.

### B. Research Questions and Contributions

We address three questions: (1) how do temporal, spatial and property patterns inform preprocessing and rent prediction; (2) what additional predictive value do contextual and historical-rent features provide beyond the core attributes; and (3) which evaluated model performs most reliably on later observations, and where do its errors remain largest?

Our work connects exploratory analysis to explicit preprocessing decisions, compares alternative treatments of repeated records, and evaluates Ridge regression, CatBoost and a multilayer perceptron under chronological validation. Feature comparisons retain the same individual validation records and original rents. Data-dependent transformations and rental-history summaries are constructed from the training portion of each experiment. The selected model is assessed through later-period errors and a longer forecasting-horizon diagnostic, rather than through an aggregate score alone. These contributions concern the application and evaluation of existing methods; their value lies in identifying which choices support reliable prediction for this dataset.

## II. Data and Problem Formulation

### A. Prediction Task and Core Data

The CS5228 competition supplies labelled training records and an unlabelled test set [1]. We learn a regression function $\hat{y} = f(x)$, where $y$ is `MONTHLY_RENT` in Singapore dollars (SGD). The ten supplied predictors describe the approval

month, town, block, street, flat type, flat model, floor area, furnishing status, lease commencement year and upfront fee. All 150,000 training rows are retained; preprocessing is detailed in Section III.

TABLE I
OVERVIEW OF THE SUPPLIED RENTAL DATA.

| Property | Training data | Test data |
|---|---|---|
| Raw records | 150,000 | 50,000 |
| Rental months | Jan. 2021–Mar. 2025 | Mar. 2025–Jul. 2026 |
| Supplied predictors | 10 | 10 |
| Target labels | Available | Unavailable |
| Final modelling records | 150,000 | 50,000 |

As Table I shows, the split is predominantly temporal: only 380 test records share March 2025 with the training period; all others are later. The maximum test horizon is 16 months beyond the last training month. We assess prediction error primarily with root mean squared error (RMSE) in SGD; the evaluation protocol is specified in Section V.

### B. Auxiliary Data Sources

The course supplied six auxiliary datasets: HDB block metadata, Mass Rapid Transit (MRT) stations, schools, shopping malls, Certificate of Entitlement (COE) auction prices, and stock prices for Singapore companies. Building attributes and coordinates support address joins and spatial features; the COE and stock tables contain dated market observations.

Our analyses used the supplied CSV files. The COE explanation [2] and company-ranking page [3] linked in the project description provide background rather than additional data collected for this study. Section III-F explains how we processed these sources and decided which features to retain.

## III. Exploratory Data Analysis and Preprocessing

### A. Target Distribution and Data Quality

The 150,000 raw training records contain no missing cells. Monthly rent has a mean of SGD 2,719, a median of SGD 2,700, and a range of SGD 300 to SGD 7,600. Its skewness is 0.39. The interquartile-range rule flags 752 records (0.50%) as extreme, but the supplied attributes cannot establish that these are recording errors. We retained them and used all original rents for evaluation. Ridge and CatBoost learned from raw SGD targets; the MLP standardised targets within each training fold and converted predictions back to SGD. No logarithmic target transformation was applied.

Two predictors have no variation: `FURNISHED` is always `yes`, and `FEE` is always zero. We removed both. Raw completeness does not eliminate the need for missing-value handling: masking invalid ages and cleaning sentinels in the auxiliary tables introduce missing values later in the pipeline.

### B. Temporal Patterns and Distribution Shift

Fig. 1 shows rapid rent growth followed by a flatter period. Mean rent rises from SGD 2,121 in 2021 to SGD 3,036 in 2023, then changes less in 2024 (SGD 3,110) and January to March 2025 (SGD 3,142). The last value covers only three months. The average of the first six monthly means is SGD 2,079, compared with SGD 3,137 for the final six. These changes motivate a continuous month index and calendar features, but also make extrapolating an earlier growth rate a potential source of error. We therefore trained on earlier months and evaluated on later months, as specified in Section V.

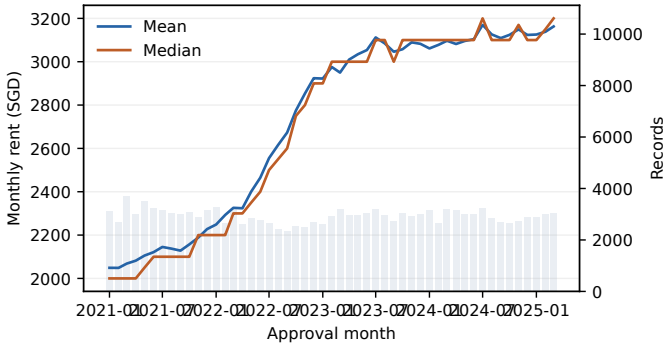



Fig. 1. Monthly mean and median rent across all 150,000 raw training records. Faint bars show record counts on the right axis. Early 2025 covers January to March only.

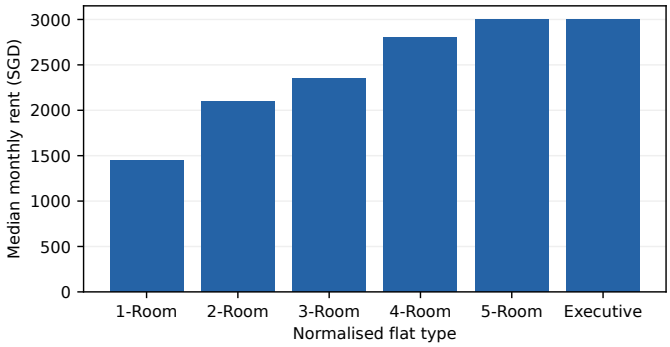



Fig. 2. Median rent by normalised flat type across the raw training records. The comparison does not adjust for month, location, or floor area.

### C. Spatial and Property Patterns

Rent varies across both property types and towns. After normalising flat-type labels, median rent increases from SGD 2,100 for two-room flats to SGD 3,000 for five-room and executive flats (Fig. 2). Floor area has a positive raw correlation with rent (Pearson $r = 0.33$), although it is closely related to flat type. We retained both descriptors. Town comparisons also differ with size: Central has a median rent per square metre of SGD 41.10, compared with SGD 23.44 in Pasir Ris, whose mean floor area is larger. These comparisons pool different months and property mixes; they support location features without identifying causal location premiums. Rent per square metre was used only for EDA because it contains the target.

### D. Core Cleaning and Feature Construction

We applied Unicode NFKC normalisation, lowercasing, and whitespace cleanup to categorical fields, removed spaces within block identifiers, and merged equivalent labels such as `4 room` and `4-room`. Flat-type cardinality fell from eleven to six, and street cardinality from 1,146 to 576. Space-separated room labels occur mainly in 2021, so leaving the spellings separate would mix property categories with formatting changes over time.

Approval month was parsed into year, month, quarter, a month index, and sine/cosine seasonal terms. We added town/street and town/block/street keys for address resolution, an ordered flat-type code, lease age, its invalidity flag, and $\log(1 + \text{floor area})$ alongside raw area. These transformations give 19 core predictors. Original row IDs and parsed dates remain alignment and validation fields, not model inputs.

We also tested four rental-history fields derived from the labelled core records: past mean rent and observation count for town/flat-type pairs and for properties. Training histories exclude the current month and all later months. Validation and test histories use only known training rents, with summaries held fixed throughout each prediction window. These are candidate features derived from rental records, not external auxiliary data.

Table II summarises the cleaning decisions. We checked the uncertain choices through the preprocessing audit below rather than assuming that fewer records or more elaborate imputation would improve prediction.

TABLE II
EDA FINDINGS AND THE CORRESPONDING PREPROCESSING DECISIONS.

| Finding | Decision and justification |
|---|---|
| Constant fields | Remove fields without predictive variation. |
| Inconsistent category labels | Normalise case, whitespace and equivalent room labels. |
| Changing rent trend | Encode approval time and validate on later months. |
| Invalid ages and sentinels | Mask source values, attempt training-only address recovery, and retain quality flags. |
| Identical observations | Retain all rows; exact removal gave negligible benefit in controlled validation. |
| Extreme targets | Retain rents flagged by IQR without evidence of error. |

### E. Preprocessing Audit and Policy Selection

The preprocessing audit compared three record policies and two missing-value treatments using Ridge with $\alpha = 10$ and the same 35 core/HDB/MRT features. Selection used July to December 2023 and January to June 2024, training on earlier months in each case. Record policies changed the training data only. The two validation blocks retained all 17,900 and 18,179 original observations, respectively, and their individual rent labels. Table III reports the unweighted mean RMSE across these two windows.

*1) Repeated observations:* The raw data contain 425 exact matches beyond the first occurrence; normalisation raises this count to 641. Unit numbers are unavailable, so identical attributes and rent cannot establish that a transaction was recorded twice. Multiple units in the same block can share the supplied description. The audit also found 21,974 repeated feature rows when rent was excluded from the grouping key. The largest group contains ten observations, and 17,900 feature groups have more than one rent value. These counts distinguish identical complete records from repeated descriptions with different targets.

We tested retaining all rows, removing exact matches including rent, and collapsing each identical feature vector to one row with mean rent. The last policy reduces a group's contribution to training because no group-size weight is retained. Its mean RMSE was about 603.96 SGD, compared with about 600.80 SGD for the other policies. We did not average validation labels under any policy.

TABLE III
TRAINING-POLICY AND MISSING-VALUE AUDIT. MEAN RMSE IN SGD OVER THE TWO EARLIER SELECTION WINDOWS; VALIDATION OBSERVATIONS ARE IDENTICAL FOR ALL ROWS.

| Training record policy | Median only | Mask/recover |
|---|---|---|
| Retain all | 600.803 | 600.808 |
| Remove exact matches | 600.798 | 600.792 |
| Collapse to mean rent | 603.965 | 603.962 |

*2) Invalid values and address recovery:* Lease age is approval year minus lease commencement year. Ages below zero or above 100 years are invalid under our rule. One training record is affected: a November 2021 approval at Bedok block 153B, Bedok South Road, has lease commencement year 2022. The masking/recovery variant removes this source value as well as its derived age. HDB maximum floors and completion years at or below zero are also masked, and completion years are checked for an implied age outside the same range. Each HDB source field is invalid or missing for 159 training records across eleven addresses. None of these audited numeric fields is invalid or missing in the supplied test rows. These checks identify values the model should not use directly; they do not determine corrected values.

Recovery uses the full normalised town/block/street key. A source attribute can be recovered only when all valid training values at that address agree. Conflicting values and unseen addresses retain missing values. We fit the lookup within each training fold, recheck recovered years against the approval year, recompute ages, and keep the original invalidity flags. Remaining numeric gaps use training-fold medians for Ridge and MLP, or CatBoost's native missing-value handling.

The recovery audit filled no source values in either selection window. In the first validation block, nine records lacked maximum-floor values and nine lacked completion years; all remained missing. The second block had no missing audited source attributes. Since HDB attributes come from a shared block lookup, another transaction at the same address need not supply an independent valid value. The result gives no

evidence of a predictive gain from recovering attributes in these windows.

The selection rule preferred retaining records when its mean RMSE was within 1 SGD of the best policy, and preferred masking/recovery when within 0.1 SGD of the best missing-value treatment. Retention with masking/recovery scored 600.808 SGD, only 0.016 SGD above exact removal with the same treatment. Its score was also 0.005 SGD above retention with median-only handling. We selected it to preserve potentially valid transactions and mask invalid source years, rather than claiming an accuracy improvement from imputation. These tolerances are practical selection rules, not significance thresholds. The final pipeline therefore retains all 150,000 training records and every one of the 50,000 test IDs.

### F. Auxiliary Construction and Record Integrity

The 9,464 HDB block records have unique normalised town, block, and street keys, with 100% address coverage in train and test. A many-to-one join adds coordinates, maximum floor, completion year, postal sector, subzone, planning area, region, and derived age/quality fields. The eleven blocks with invalid floor/year sentinels account for the 159 affected training records; 285 block postal codes marked `NIL` become missing categories.

Rail data provide current status but no opening dates. We used a fixed proxy of older line prefixes, excluded eight specified later/future codes, and deduplicated station names and coordinates, leaving 122 locations. The proxy excludes TEL and LRT and does not track openings during the rental period. Haversine distances yield the nearest-station distance and counts within 500 metres and one kilometre. School and mall distances/counts were built similarly and tested as separate feature groups. They were excluded from the selected configuration; their snapshots also lack opening histories.

We cleaned the COE and stock observations, aggregated them by calendar month, and applied a one-month lag. Although the supplied files include market observations covering test-period rental months, our forecasting evaluation uses a fixed prediction cutoff. Later market observations are not treated as information known at that cutoff. A lagged feature for a distant forecast can still require these observations, so both market feature groups were excluded from the final model.

The base configuration contains 35 predictors: 19 core fields and sixteen HDB/MRT fields. Adding the four rental-history fields described in Section III-D gives a 39-input candidate. The selected model retains the 35-input base configuration, as specified in Section IV-D. Joins check key uniqueness, row counts, and matching IDs/dates so that feature construction preserves the evaluation population.

## IV. Data Mining Methods and Model Development

### A. Reference Predictor and Ridge Baseline

The initial reference predictor used the historical median rent for each town/flat-type pair, falling back to the town median, flat-type median, and overall training median when necessary. This provides a simple location/size reference but does not explicitly represent market changes. Its original evaluation removed exact duplicate records, so its scores are not included in the common-population comparison below.

Ridge regression served as the main reference for the improvement experiments. It minimises squared prediction error with an $L_2$ coefficient penalty, which helps control correlated inputs [4]. Its sparse one-hot representation makes it inexpensive to fit and useful for checking preprocessing and feature choices. The baseline used 35 core/HDB/MRT predictors, standardised numeric inputs, and untransformed SGD rents. We selected $\alpha = 10$ over 100 using the earlier validation windows. Fitting used the LSQR solver with tolerance $10^{-4}$ and at most 10,000 iterations.

### B. Candidate Models

*1) CatBoost:* We evaluated gradient-boosted trees to represent nonlinear relationships and interactions among time, location and property descriptors. CatBoost accepts categorical fields directly, avoiding the external one-hot pipeline used by Ridge [5]. We fitted raw SGD targets with RMSE loss, seed 42, and four CPU threads. Input rows were chronological and `has_time=True` preserved their order during categorical processing and tree construction [6]. Numeric missing values were handled natively. Tree count was chosen using a chronological block inside the training data, with a 500-tree budget and 40-round early stopping.

*2) Multilayer perceptron:* The MLP used two hidden layers with 64 and 32 ReLU units, Adam optimisation, $L_2$ regularisation 0.01, initial learning rate 0.001, and batch size 1,024. Numeric inputs were standardised and categorical inputs were sparse one-hot vectors. Training rents were standardised and predictions converted back to SGD. We used `MLPRegressor` with incremental epoch-wise training and disabled its random validation split [7]. Internal chronological RMSE selected the epoch count, with a maximum of 40 epochs; training stopped after six epochs without an improvement exceeding 0.1 SGD. The chosen duration was then used to refit on the complete outer training portion. Seeds 42, 123 and 2026 assessed stability; seed 42 was fixed for model selection. Architecture, learning rate and regularisation were not grid-searched. The MLP used the 35 base predictors without feature-group tests.

### C. Completed Model Improvements

The preprocessing audit retained original transactions and masked invalid source attributes as described in Section III-E. Subsequent experiments held this treatment fixed to isolate model and feature changes. Replacing Ridge with CatBoost allowed nonlinear combinations of the existing inputs. No capped trend, recency weighting, rolling training window, post-hoc calibration or ensemble was evaluated; temporal differences below describe the fitted models rather than a separate calibration method.

*1) Auxiliary and historical information:* Ridge and CatBoost each tested six school features, three mall features, their nine-feature combination, and four historical-rent summaries.

Each group was added separately to the 35 base predictors. Historical inputs were past mean rent and count by town/flat type and by property address. Training histories excluded the current month; evaluation histories were frozen at the training cutoff. Unseen groups received count zero and a missing mean, handled by the model's usual missing-value treatment. This allowed target-derived information without using evaluation rents.

*2) Tuning and selection:* We first selected model parameters on base features, then evaluated the additional groups with those parameters fixed. The expanded CatBoost grid contained 20 configurations: depths $\{6, 8\}$, learning rates $\{0.001, 0.005, 0.01, 0.05, 0.08\}$, and leaf regularisation $\{3, 5\}$. An added group was retained only if the best group reduced mean validation RMSE by at least 1 SGD relative to base features. With the selected tree parameters, history reduced RMSE by 0.889 SGD, so the simpler base feature set was retained. This tolerance expresses a practical preference for simplicity, not a statistical significance test.

### D. Final Model Specification

The earlier-fold selection chose a single CatBoost model with 35 core/HDB/MRT predictors, depth 8, learning rate 0.05, and leaf regularisation 3. It retained all training records and used invalid-source masking/address recovery. School, mall and rental-history additions were excluded. Targets remained in SGD, and no prediction clipping or other post-processing was applied in the evaluation pipeline. Internal validation selected 271 and 350 trees in the two selection folds, and 468 trees for the later diagnostic block. For the final submission, the internal split used January–March 2025 to select 499 trees from a model trained on earlier months. We then fitted a fresh 499-tree model on all 150,000 labelled records through March 2025 and predicted the 50,000 test records. The final feature contract contains 35 predictors, including eleven categorical fields; row IDs and parsed dates serve only alignment and validation.

## V. Experimental Setup

### A. Chronological Validation

Random splitting could mix later market conditions into training. Instead, we used expanding training windows and evaluated complete later months (Table IV). There was no gap between training and evaluation. Preprocessing, parameter, feature and model selection used only the two six-month windows. The selected configurations were frozen before evaluating the nine-month diagnostic block. That block was already examined during the original baseline work and is therefore not a completely untouched holdout, although its new scores did not trigger model switching.

TABLE IV
Chronological evaluation. Counts refer to original labelled records.

| Use | Evaluation months | Train | Evaluate |
|---|---|---|---|
| Selection 1 | Jul.–Dec. 2023 | 87,937 | 17,900 |
| Selection 2 | Jan.–Jun. 2024 | 105,837 | 18,179 |
| Diagnostic | Jul. 2024–Mar. 2025 | 124,016 | 25,984 |
| Long horizon | Jul. 2023–Oct. 2024 | 87,937 | 47,528 |

The separate long-horizon diagnostic trained through June 2023 and predicted 16 months without updating from evaluation rents. This matches the maximum future test horizon in duration, but covers different market conditions. It overlaps the selection and later diagnostic windows and is neither independent evidence nor an additional tuning window.

### B. Training-Fold Preprocessing and Leakage Controls

All data-dependent transformations were fitted within the relevant training portion. Ridge and MLP used numeric median imputation and standardisation, categorical mode imputation, and one-hot encoding with minimum frequency 10. Unknown categories mapped to an existing infrequent category, or to an all-zero encoding when that bucket was absent. CatBoost received normalised categorical strings directly and retained numeric gaps for native handling. Address-recovery maps and rental summaries were rebuilt for each fold. The last three training months formed the internal block for tree-count and MLP-epoch selection; its preprocessing was fitted on the preceding months. Models were then refitted on the complete outer training portion. Evaluation labels never entered these transformations or iteration selection.

### C. Hyperparameter Selection and Reproducibility

Selection minimised the unweighted mean RMSE of the two earlier windows. Ridge evaluated $\alpha \in \{10, 100\}$; CatBoost evaluated the 20-setting grid before four feature additions. Each configuration was scored on both selection windows, giving 40 base-feature CatBoost fits and eight feature-addition fits, each with its own internal tree-count selection. Ridge used four base-feature fits and eight feature-addition fits. The MLP evaluated one configuration at three seeds on both windows, giving six fits, with the predeclared seed 42 used for comparison. These unequal tuning budgets limit conclusions about model families, particularly neural networks. The recorded seed-42 MLP selected one and two epochs in the earlier folds and five epochs in the later diagnostic.

The preprocessing audits, feature comparisons, model training and final evaluation use the shared implementation in `experiments/rental_experiments.py`. CSV scores and JSON selections record fold sizes, parameters, selected iterations and runtime. The saved refit environment records Python 3.11.5, NumPy 1.26.4, pandas 2.1.4, scikit-learn 1.2.2 and CatBoost 1.2.10, together with SHA-256 hashes of the raw training data, test data and shared implementation. The current refit metadata and processed-data contract agree with the selected model; the recorded hashes match the repository inputs and implementation. The final full-data fit and test

prediction took approximately 81.4 seconds. Earlier-fold base fits took approximately 1.2–1.5 seconds for selected Ridge, 25.9–38.6 seconds for selected CatBoost, and 5.2–8.5 seconds for seed-42 MLP; these are observed pipeline runtimes rather than hardware-normalised benchmarks.

### D. Metrics and Controlled Comparisons

For original rents $y_i$ and predictions $\hat{y}_i$, we report RMSE, the square root of mean squared prediction error [8], and MAE:

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}, \quad \mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|. \quad (1)$$

RMSE is the selection metric; MAE, $R^2$ and signed bias support interpretation. Residuals are actual minus predicted, so positive mean bias indicates underprediction. Selection scores are unweighted averages of fold RMSEs, whereas later and long-horizon scores are pooled within their respective blocks. We calculated later-block $R^2$ from the saved predictions. Every new comparison used identical original validation IDs and individual rents. The initial baseline's deduplicated population is excluded from these common-population tables. We did not estimate confidence intervals or claim statistical significance from small score differences.

## VI. Final Results and Discussion

### A. Overall Model Comparison

Table V compares the frozen configurations on 25,984 later observations. CatBoost achieved RMSE 458.740 SGD versus 582.264 SGD for Ridge, a 21.21% reduction, and reduced MAE from 432.274 to 334.412 SGD. Its earlier-fold mean RMSE was also lowest: 547.372 SGD, compared with 600.808 for Ridge and 609.986 for seed-42 MLP. These earlier scores determined the winner. The MLP's later RMSE of 469.729 SGD was close to CatBoost despite its weaker earlier result; this change in ranking cautions against judging models from one period alone.

TABLE V
July 2024–March 2025 comparison (25,984 original rows). All models use 35 base features. Errors and bias are in SGD.

| Method | RMSE | MAE | $R^2$ | Bias |
|---|---|---|---|---|
| Ridge | 582.264 | 432.274 | 0.201 | −337.793 |
| MLP | 469.729 | 347.440 | 0.480 | 34.853 |
| CatBoost | 458.740 | 334.412 | 0.504 | 13.551 |

### B. Performance across Time Periods

Table VI shows that CatBoost improved on Ridge in both selection periods and the later diagnostic. Ridge's RMSE was similar across these periods, but its earlier-fold bias changed from $-72.7$ to $-300.3$ SGD, consistent with excessive extrapolation as rent growth flattened. CatBoost's later-period improvement coincided with much smaller aggregate bias. This is observed calibration behaviour, not evidence that a particular input or mechanism caused the difference.

TABLE VI
RMSE (SGD) by evaluation period. The final row overlaps the others.

| Period | Ridge | MLP | CatBoost |
|---|---|---|---|
| Jul.–Dec. 2023 | 597.807 | 618.967 | 578.121 |
| Jan.–Jun. 2024 | 603.809 | 601.005 | 516.622 |
| Jul. 2024–Mar. 2025 | 582.264 | 469.729 | 458.740 |
| 16-month diagnostic | 624.419 | — | 534.106 |

At the earlier forecast origin, CatBoost's 16-month RMSE was 534.106 SGD versus 624.419 SGD for Ridge, a 14.46% reduction. Its MAE was 410.923 SGD and bias 30.371 SGD, compared with Ridge's MAE 473.995 SGD and bias $-263.517$ SGD. The gain persists in this longer diagnostic, but a different origin and overlapping observations prevent treating it as an independent estimate of future competition accuracy.

### C. Controlled Feature Comparisons

Table VII reports completed feature additions with selected parameters held fixed. Ridge gained no useful improvement from any group. For CatBoost, schools and malls separately increased mean RMSE, their combination reduced it by 0.387 SGD, and history reduced it by 0.889 SGD. Neither reduction met the 1 SGD retention rule. Consequently, the final selection uses the existing HDB/MRT inputs without these additions. The comparison demonstrates that additional predictors do not automatically improve forecasting. It does not isolate HDB/MRT's contribution to CatBoost: no core-only CatBoost removal experiment was completed.

TABLE VII
Earlier-fold mean RMSE (SGD) at fixed selected parameters. CatBoost uses depth 8, learning rate 0.05, regularisation 3; Ridge uses $\alpha = 10$. Each addition is separate.

| Feature configuration | Ridge | CatBoost |
|---|---|---|
| Base (35 inputs) | 600.808 | 547.372 |
| Base + schools | 600.887 | 548.135 |
| Base + malls | 600.853 | 547.653 |
| Base + schools and malls | 600.885 | 546.985 |
| Base + history | 603.998 | 546.482 |

### D. Error Analysis and Model Interpretation

Ridge overpredicted later-block rent by 337.8 SGD on average, with monthly bias between $-422.4$ and $-183.8$ SGD. CatBoost's overall underprediction was 13.6 SGD, with monthly bias between $-12.2$ and 39.3 SGD. Thus, its aggregate RMSE gain accompanied substantially better temporal calibration.

Errors still varied across subgroups. CatBoost RMSE was 432.5 SGD for three-room flats (8,562 rows), 462.6 for four-room flats (9,468), 469.4 for five-room flats (5,927), and 551.0 for executive flats (1,309). Executive flats were underpredicted by 40.7 SGD on average. Only ten one-room rentals were available, so their subgroup score is too sparse for a broad conclusion. Among towns with at least 200 observations, Jurong West had the largest RMSE, 561.0 SGD (1,802 rows), and bias 83.1 SGD. Properties with fewer than ten training rentals had

RMSE 477.9 SGD (5,447 evaluation rows), compared with 453.5 SGD for more frequently observed properties (20,537 rows). These comparisons describe observed groups without controlling their other characteristics.

The largest saved CatBoost error concerned a five-room Bukit Merah rental in March 2025: actual rent was 650 SGD and prediction approximately 3,858 SGD, an overprediction of 3,208 SGD. The supplied fields cannot establish whether such an unusually low rent reflects contract terms, flat condition or recording error. It remains in evaluation. No feature-attribution analysis was completed, so model improvements cannot be assigned to causal location or amenity effects.

### *E. Test Predictions and Submission Status*

The final `submissions/catboost_improved.csv` contains 50,000 finite predictions in exactly the sample-submission ID order. It was produced by the selected 35-input CatBoost configuration refitted with 499 trees on all 150,000 labelled records. The saved `final_refit.json` and `data/processed/improved/preprocessing_metadata.json` document the refit and feature contract. Processed training and test files preserve all records; these full-data artifacts are for final prediction, while validation rebuilds data-dependent transformations inside each fold. No verified competition score is available in the experiment artifacts, so the reported accuracy remains internal chronological performance.

### *F. Limitations of the Findings*

Unit-level attributes and contract details are unavailable, leaving substantial unexplained variation and uncertainty about repeated records. Amenity snapshots lack complete opening histories, while the MRT proxy does not reconstruct station availability month by month. Historical summaries are frozen during forecasting and can become stale. CatBoost's long-horizon gain does not guarantee extrapolation under future market changes.

Model and feature selection reused two windows for multiple comparisons, and the later block had already been inspected in the baseline work. The overlapping long-horizon diagnostic adds a different forecasting origin, not independent replication. The 500-tree cap also constrains the CatBoost comparison: all configurations with learning rates 0.001, 0.005 and 0.01 used the full budget in both selection windows. Their higher errors may partly reflect insufficient iterations rather than intrinsically unsuitable learning rates. Parameters were selected on base features before feature additions, so the experiment does not exhaustively search parameter/feature combinations.

MLP results reflect a fixed architecture and limited optimisation checks; mean RMSE varied from 609.986 to 639.407 SGD across seeds. They do not establish a general disadvantage of neural networks. The final submission matches the selected configuration, but future test labels remain unavailable; its realised accuracy cannot be inferred from file integrity or the internal diagnostic scores.

## VII. Conclusion

Our analysis connected changing rental levels, inconsistent category labels and uncertain repeated records to explicit preprocessing and validation choices. Retaining potentially legitimate observations and masking invalid source attributes preserved the evaluation population; the audit did not show a meaningful accuracy gain from address recovery. Among the evaluated models, CatBoost with 35 core/HDB/MRT predictors performed best on the two selection windows and reduced later-block RMSE from 582.264 to 458.740 SGD relative to Ridge, with substantially smaller signed bias. Additional school, mall and historical-rent features did not meet the 1 SGD retention threshold, so most of the measured improvement came from changing the model. The longer diagnostic supported this result across a 16-month horizon, but overlapped the other evaluation windows. Larger errors for executive flats and sparsely observed properties, incomplete amenity histories and unavailable future test labels constrain generalisation. The selected model provides a useful attribute-based rental reference whose accuracy must be assessed across periods and subgroups as market conditions change.

## REFERENCES


[1] National University of Singapore, "CS5228 course project," Kaggle competition and supplied rental and auxiliary datasets. [Online]. Available: https://www.kaggle.com/competitions/cs-5228-2610-project

[2] Wikipedia contributors, "Certificate of Entitlement," *Wikipedia*. [Online]. Available: https://en.wikipedia.org/wiki/Certificate_of_Entitlement. Accessed: Oct. 10, 2026.

[3] CompaniesMarketCap, "Largest companies in Singapore by market capitalization." [Online]. Available: https://companiesmarketcap.com/sgd/singapore/largest-companies-in-singapore-by-market-cap/. Accessed: Oct. 10, 2026.

[4] scikit-learn developers, "Ridge." [Online]. Available: https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.Ridge.html. Accessed: Oct. 10, 2026.

[5] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin, "CatBoost: unbiased boosting with categorical features," in *Advances in Neural Information Processing Systems*, 2018. [Online]. Available: https://arxiv.org/abs/1706.09516.

[6] CatBoost developers, "Common parameters." [Online]. Available: https://catboost.ai/docs/en/references/training-parameters/common. Accessed: Oct. 10, 2026.

[7] scikit-learn developers, "MLPRegressor." [Online]. Available: https://scikit-learn.org/stable/modules/generated/sklearn.neural_network.MLPRegressor.html. Accessed: Oct. 10, 2026.

[8] Statistics How To, "RMSE: Root Mean Square Error." [Online]. Available: https://www.statisticshowto.com/probability-and-statistics/regression-analysis/rmse-root-mean-square-error/. Accessed: Oct. 10, 2026.


## AI DECLARATION

We used GPT-6-Sol, GPT-6-Astra and GPT-6.1-Sol for file formatting, grammar correction and put up README descriptions for our notebooks. We declare our ideas, analysis and model selections are original.

## APPENDIX A
## TEAM WORKLOAD BREAKDOWN

*Writing prompt: Record actual completed contributions. The names, IDs and team name come from the supplied progress slides. Confirm them before submission. Only Sun Zekun's core-preprocessing responsibility is prefilled from the conversation; complete all other entries from the team's actual workload.*

TABLE VIII
TEAM WORKLOAD AND REPORT CONTRIBUTIONS. STUDENT IDS APPEAR IN THE AUTHOR BLOCK.

| Member | Project responsibilities | Report contributions |
|---|---|---|
| Li Chengxuan | *TBD* | *TBD* |
| Mao Huanqing | *TBD* | *TBD* |
| Sun Zekun | Core data preprocessing; other tasks *TBD* | Core preprocessing; other tasks *TBD* |
| Zhang Ziqing | *TBD* | *TBD* |

## APPENDIX B
## SUPPLEMENTARY DETAILS

*Writing prompt: Use sparingly. Include only the full hyperparameter search/configuration table, a compact feature dictionary, or supplementary EDA/error tables that help reproduce or elaborate the main results. Do not move the final model definition, validation design, main results or essential interpretation out of the main text. The paper must remain comprehensible without this appendix.*